Statistical Analysis of Software Bug Resolution Times

Introduction

This section examines software bug resolution times to assess the suitability of a lognormal model. Graphical summaries, log transformation, maximum likelihood estimation, and a chi-square goodness-of-fit test are used to evaluate the distribution.

1. Data Description and Initial Analysis

1.1 Data Visualization

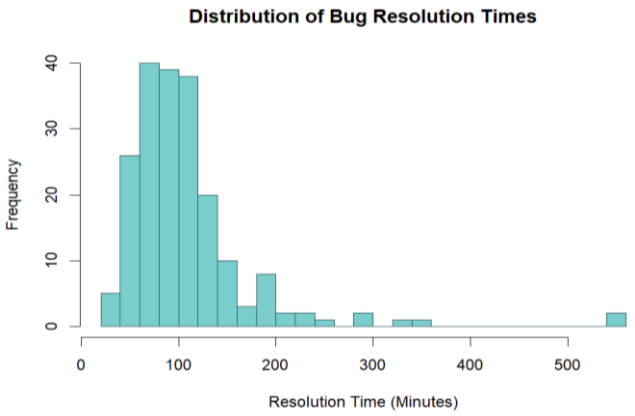

Figure 1: Histogram of Bug Resolution Times

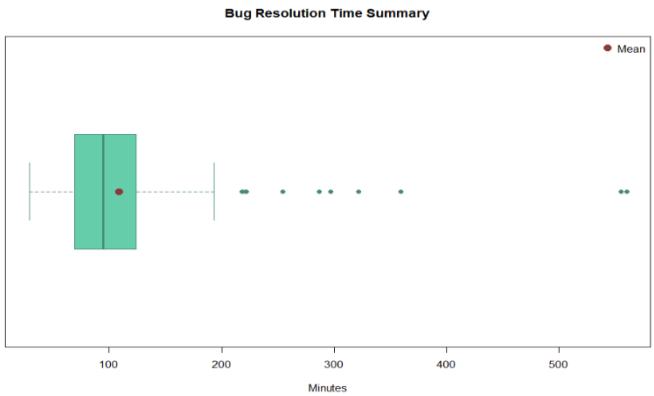

Figure 2: Boxplot of Bug Resolution Time

Figure 1 presents a histogram of the raw bug resolution times. The distribution displays a clear right skew, with most resolution times concentrated between 50-150 minutes but with several extreme values extending beyond 500 minutes. This skewness is typical for this data, where most issues are resolved relatively quickly, but some complex bugs require substantially more time.

Figure 2 shows a boxplot summary of the resolution times, with a red dot indicating the sample mean (108.55) confirms the right skewness as it is greater than the median (95) observed in the histogram. The median is closer to the first quartile (69.75) than the third quartile (123.5) which also proves the right skew. Several outliers above 200 minutes, with one extreme value at 560 minutes.

1.2 Numerical Summaries

Min	Q1	Median	Q3	Max	Mean	SD	IQR
29	69.75	95	123.5	560	108.55	68.12	53.75

Table 1: Summary Statistics for Bug Resolution Times

In **Table 1**, the substantial difference between the mean and median (13.55 minutes), combined with the standard deviation being approximately 63% of the mean, indicates considerable variability and non-normality in the raw data.

2. Log Transformation and Parameter Estimation

2.1 Maximum Likelihood Estimation Derivation

Let $Y_i = \ln(X_i)$ represent the natural logarithm of bug resolution times, where X_i are the original resolution times. Assuming $Y_i \sim N(\mu, \sigma^2)$, the likelihood function is:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(y_i-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{(-n/2)} e^{\frac{-1}{(2\sigma^2)} \sum_{i=1}^{n}(y_i-\mu)^2}$$

The log-likelihood function is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2$$

Taking partial derivatives:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - \mu) = 0$$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(y_i - \mu)^2 = 0$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

2.2 Numerical Parameter Estimates

$\hat{\mu}$	$\hat{\sigma}^2$	s^2
4.5634	0.2187	0.2198

Table 2: Log-Normal Parameter Estimates

As seen in **Table 2** ($Y = \ln(\text{bug_time})$). The MLE of μ is identical to the sample mean, as expected from the derivation. The MLE of σ^2 (0.2187) differs slightly from the sample variance (0.2198) because the sample variance uses $n - 1$ in the denominator while the MLE uses n. For $n = 200$, this difference is minimal, confirming that both estimators are consistent and asymptotically equivalent.

3. Normality Assessment of Log-Transformed Data

Figure 3 presents a Q-Q plot of the log-transformed bug resolution times against a theoretical normal distribution. Most points follow the reference line closely. Minor deviations occur in the tails, which is common even for normally distributed data.

No curvature or pronounced S-shape pattern, suggesting the log-transformed data approximately follows a normal distribution. This supports the hypothesis that the natural logarithm of bug resolution times can be reasonably modelled by a normal distribution, justifying the use of lognormal distribution for the original data.

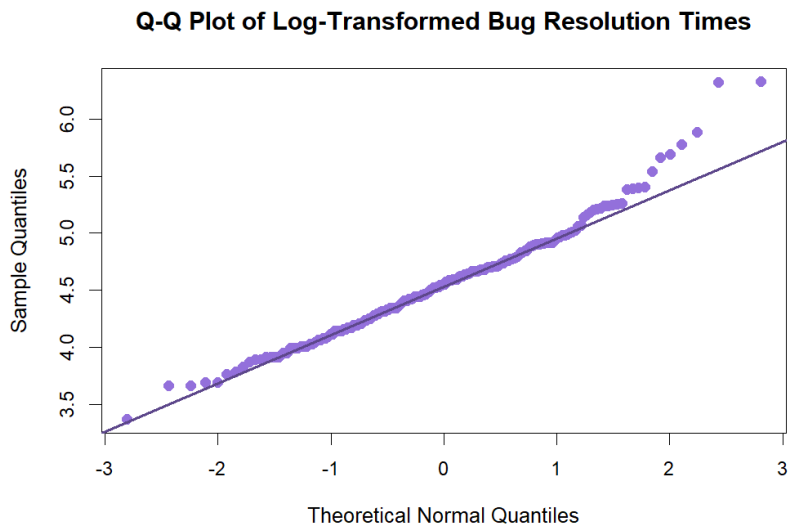

Figure 3: Q-Q Plot of Bug Resolution Times against normal quantiles

4. Goodness-of-Fit Test for Lognormal Distribution

4.1 Hypotheses

- **Null Hypothesis (H_0):** The bug resolution times follow a lognormal distribution with parameters $\mu = 4.5634$ and $\sigma^2 = 0.2187$
- **Alternative Hypothesis (H_1):** The bug resolution times do not follow the specified lognormal distribution

4.2 Test Procedure

We performed a χ^2 goodness-of-fit test with 10 equiprobable cells based on the lognormal quantiles:

Breakpoints: $0.00, 52.67, 64.70, 75.05, 85.19, 95.91, 107.97, 122.56, 142.16, 174.64, \infty$

4.3 Test Results

Breakpoints	0.00 – 52.67	52.67 – 64.70	64.70 – 75.05	75.05 – 85.19	85.19 – 95.91	95.91 – 107.97	107.97 – 122.56	122.56 – 142.16	142.16 – 174.64	174.64 - ∞
Observed	17	22	22	23	17	24	24	19	12	20
Expected	20	20	20	20	20	20	20	20	20	20

Table 3: Chi-Square Goodness-of-Fit Test Results

As seen in **Table 3**, expected frequencies: 20 in each cell (200 × 0.10). Degrees of freedom: $df = 10 - 1 - 2 = 7$ (10 cells - 1 for probability constraint - 2 estimated parameters).

χ^2	p-value
6.6	0.4717

Table 4: Chi-Square Test Statistics

4.4 Interpretation

The p-value of 0.4717 exceeds the significance level of 0.05. Therefore, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the bug resolution times do not follow a lognormal distribution with the estimated parameters.

The test statistic (6.6) is well below the critical value of $\chi^2_{0.05,7} = 14.07$, and the observed frequencies show reasonable agreement with the expected frequencies across all cells.

5. Conclusion

The analysis reveals that bug resolution times exhibit right-skewness with occasional extreme values, a pattern typical for time-to-resolution data. The natural logarithm transformation successfully normalizes the data, as evidenced by the Q-Q plot showing approximate linearity. The lognormal distribution provides a statistically adequate fit, supported by both visual assessment and formal hypothesis testing (χ^2 goodness-of-fit test, p=0.472).

These findings have practical implications for the company's bug management processes. The lognormal model can be used to predict resolution times and establish quality control benchmarks. Specifically, approximately 95% of bugs should be resolved within $[\mathbf{52.7}, \mathbf{174.6}]$ minutes, calculated as $\boldsymbol{e}^{(\mathbf{4.5634} \pm \mathbf{1.96} \times \sqrt{\mathbf{0.2187}})}$. This provides a solid statistical foundation for process improvement, resource planning, and setting realistic expectations in the bug management system.

Statistical Analysis of App Session Durations

Introduction

This section analyses mobile app session durations using graphical summaries and bootstrap methods for inference. Assuming a lognormal model, asymptotic and parametric bootstrap approaches are applied and compared.

1. Data Description and Initial Analysis

1.1 Data Visualization

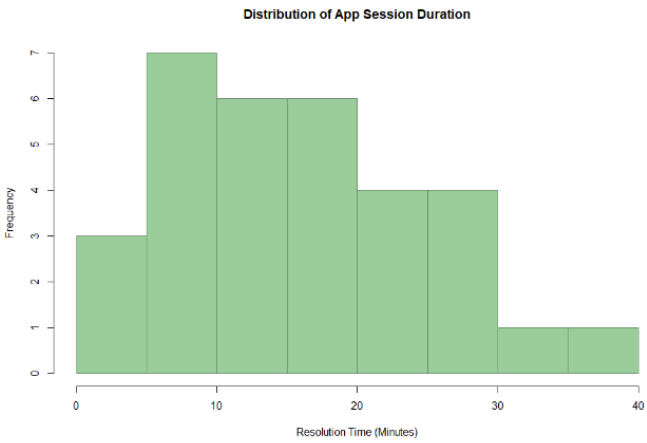

Figure 4: Histogram of App Resolution Times

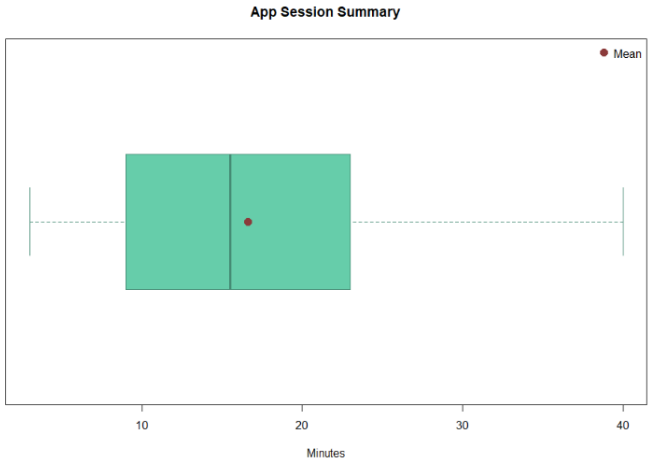

Figure 5: Boxplot of App Resolution Times

Figure 4 presents a histogram of the raw app session durations. The distribution displays moderate right skewness, with most session durations concentrated between 5-20 minutes but with some values extending up to 40 minutes. This pattern is typical for session duration data, where most users engage briefly while some exhibit extended usage.

Figure 5 shows a boxplot summary of the session durations, with a red dot indicating the sample mean. The mean (16.62 minutes) exceeding the median (15.5) and the upper whisker extending further than the lower whisker both prove the positive skew.

1.2 Numerical Summaries

Min	Q1	Median	Q3	Max	Mean	SD	IQR
3	9	15.5	22.5	40	16.62	9.33	13.5

Table 5: Summary Statistics for App Session

For the sample of 32 sessions. We can see in **Table 5**, the moderate difference between the mean and median (1.12 minutes), combined with the standard deviation being approximately 56% of the mean, indicates some variability and non-normality in the raw data. The IQR of 13.5 minutes suggests considerable spread in the middle 50% of sessions.

1.3 Distributional Implications

The observed right skewness has important implications for statistical inference. Parametric methods assuming normality may yield biased results. Transformations (logarithmic) may improve symmetry. Non-parametric methods (particularly bootstrap approaches) would help. Median-based summaries may be more representative than mean-based summaries.

2. Non-Parametric Bootstrap Analysis

2.1 Methodology

A non-parametric bootstrap procedure was implemented with $B = 100{,}000$ resamples to estimate sampling distribution of the sample median. Each bootstrap sample was generated by random sampling with replacement from the original data, maintaining the same sample size (32).

2.2 Empirical Sampling Distribution

Figure 6 presents the bootstrap distribution of the sample median. The distribution deviates from normality, showing noticeable irregularities including two prominent tall bars that create a bimodal appearance rather than a smooth bell curve.

While roughly symmetric overall, the distribution exhibits non-normal features, the two dominant bars around 15 minutes create an irregular peak structure. A Moderate spread, range spanning approximately 8 to 24 minutes. And Asymmetric tails, slight imbalance in the upper and lower tails.

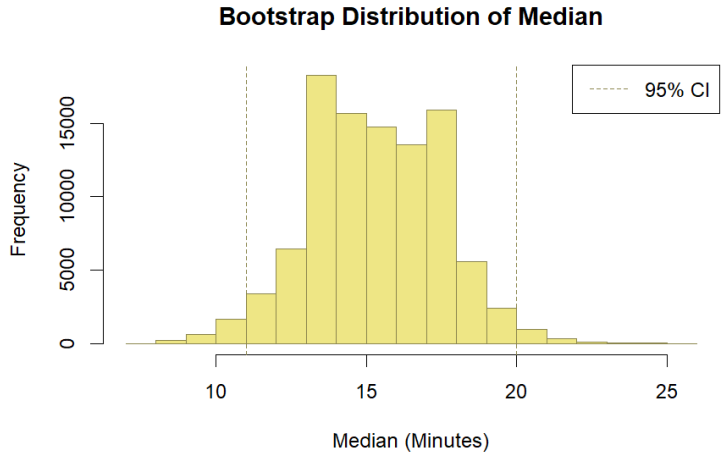

Figure 6: Histogram of Bootstrap Distribution of Median

Min	Q1	Median	Q3	Max	Mean	SD	IQR
7.5	14	15.5	17.5	26	15.61	2.2	3.5

Table 6: Numerical summary for Bootstrap distribution

Although asymptotic theory suggests that the sampling distribution of the median should approach normality for large samples, with $n = 32$ some irregularity remains visible. The bootstrap distribution of the median (100,000 resamples) retains some irregular features, possibly due to the original data's right skewness (including values up to 40 minutes) and limited sample size. Despite this, the distribution is nearly symmetric with mean 15.61 and median 15.5 minutes, indicating minimal bias. The standard deviation of 2.2 minutes reflects moderate sampling variability, while the interquartile range of 3.5 minutes shows that the middle 50% of estimates fall between 14 and 17.5 minutes, substantially smaller than the original data's IQR (13.5 minutes), demonstrating that the sampling distribution of the median is considerably less variable than the underlying population.

2.3 Confidence Interval Estimation

Lower Bound	Upper Bound	Interval width
11.5	20	8.5

Table 7: 95% non-parametric bootstrap confidence interval for the population median

We are 95% confident that the true population median duration is between 11.5 and 20 minutes mentioned in **Table 7**. This interval provides a robust estimate that does not rely on distributional assumptions.

3. Lognormal Modelling and Asymptotic Inference

3.1 Model Specification

The app session duration data was modelled using a lognormal distribution, where $if\ X \sim Lognormal(\mu, \sigma^2)$, then $Y = ln(X) \sim N(\mu, \sigma^2)$. This transformation is particularly suitable for positive, right-skewed data such as duration measurements.

3.2 Parameter Estimation

For the log-transformed data $Y = ln(session_duration)$:

$\hat{\mu}$	$\hat{\sigma}$	σ^2
2.6895	0.6006	0.3607

Table 8: Log-Normal Parameter Estimates

3.3 Asymptotic Confidence Interval

Under the lognormal model, if $X \sim \text{Lognormal}(\mu, \sigma^2)$, then the population median equals e^{μ}. Therefore, inference for the median can be conducted by first constructing a confidence interval for μ, and then exponentiating the endpoints.

Standard error of $\hat{\mu}$: $SE(\hat{\mu}) = \hat{\sigma}/\sqrt{n} = 0.6006/\sqrt{32} = 0.1062$

95% CI for μ: $[\hat{\mu} - 1.96 \times SE(\hat{\mu}), \hat{\mu} + 1.96 \times SE(\hat{\mu})] = [2.4814, 2.8976]$

95% CI for population median: $[e^{2.4814}, e^{2.8976}] = [12.37,\ 19.40]$ minutes

Aspect	Bootstrap CI	Asymptotic CI
Interval	[11, 20]	[12.37, 19.40]
Width	9 minutes	7.03 minutes
Assumptions	None	Lognormal distribution
Theoretical basis	Empirical	Asymptotic normal theory
Upper bound difference	Higher by 2.59 minutes	Lower bound nearly identical

Table 9: Method Comparison

From **Table 9**, the bootstrap CI $[11, 20]$ and asymptotic lognormal CI $[12.37,\ 19.40]$ yield similar lower bounds, but the bootstrap interval is wider, particularly at the upper bound (20 vs 19.40). The narrower asymptotic interval (7.03 vs 9.00 minutes) reflects parametric efficiency when distributional assumptions hold, while the wider bootstrap interval reflects sampling variability without distributional assumptions, which can result in larger intervals when the sampling distribution is skewed. With $n = 32$, the sample size is relatively small, so asymptotic approximations may be less precise, whereas the bootstrap method directly estimates the sampling distribution from the data.

4. Bootstrap Analysis of Interquartile Range

4.1 Confidence Interval Estimation

Lower Bound	Upper Bound	Interval width
8.25	18.5	10.00

Table 10: 95% non-parametric bootstrap confidence interval for the population IQR

4.3 Comparative Analysis

Statistic	Point Estimate	95% CI	CI Width
Median	15.50 minutes	[11, 20]	8.5 minutes
IQR	13.50 minutes	[8.25, 18.25]	10.00 minutes

Table 11: Comparing the IQR CI with the median CI reveals important

Looking at **Table 10** and **Table 11**, using $B = 100{,}000$ resamples, the 95% bootstrap CI for the population IQR is $[8.25, 18.25]$ (width of 10.00 minutes). Comparing with the median CI reveals that the IQR exhibits approximately 14% greater uncertainty (10.00 vs 8.5 minutes width). This indicates that estimating spread is more challenging than estimating central tendency with this sample size, with practical implications for resource planning where understanding variability is crucial.

5. Hypothesis Test for Median Session Duration

An analyst claims that the population median app session duration is 15 minutes or less. We assess this claim using a parametric bootstrap hypothesis test.

5.1 Hypothesis

- Null Hypothesis (H_0): Population median $\leq$ 15 minutes
- Alternative Hypothesis (H_1): Population median $>$ 15 minutes
- Significance Level: $\alpha = 0.05$

5.3 Test Procedure

A parametric bootstrap test was conducted under the assumption that session durations follow a lognormal distribution. We estimated $\hat{\sigma} = 0.6006$ from the data, set $\mu_0 = ln(15) = 2.7081$ under H_0, generated $B = 100{,}000$ bootstrap samples from Lognormal$(\mu_0, \hat{\sigma}^2)$, and computed the p-value as the proportion of bootstrap medians ≥ the observed median of 15.50 minutes.

5.4 Test Results

The bootstrap p-value was 0.410. Since this exceeds 0.05, we fail to reject the null hypothesis.

5.5 Interpretation

There is insufficient statistical evidence to contradict the analyst's claim. The observed median of 15.50 minutes could reasonably occur by chance if the true median were 15 minutes, the 0.5-minute difference is not statistically significant. Given the relatively small sample size ($n = 32$), the test may have limited ability to detect small deviations from 15 minutes. This result should be interpreted alongside the confidence interval $[11, 20]$, which suggests the true median could plausibly be below, at, or above 15 minutes.

6. Conclusions

The analysis reveals that app session durations exhibit moderate right skewness, with typical sessions lasting between 9 and 22.5 minutes (IQR). The population median is estimated to fall between 11.5 and 20 minutes (95% CI), while IQR estimation shows approximately 14% greater uncertainty than the median, indicating that characterizing spread is more challenging than central tendency with this sample size. The lognormal distribution provides a reasonable fit, supported by consistency between parametric and non-parametric methods. The analyst's claim that the median session duration is 15 minutes or less cannot be rejected ($p = 0.405$), meaning the data do not provide statistically significant evidence that the true median exceeds 15 minutes.

These findings have practical implications for understanding user engagement patterns. Bootstrap methods provide robust inference without distributional assumptions, automatically accounting for skewness in the sampling distribution, while parametric approaches offer greater efficiency (narrower confidence intervals) when distributional assumptions hold. With $n = 32$, the sample provides adequate precision for median estimation but limited precision for characterizing spread. The similarity between parametric and non-parametric results (particularly for the lower bound) suggests that the lognormal approximation is reasonable for this dataset, while their divergence at the upper bound highlights the value of distribution-free methods as sensitivity checks for resource planning and user experience optimization.

Appendix: R Code

```
bug_time = scan("C:/Users/basil/OneDrive/Desktop/Base/Studies/Undergraduate/Semester 6/F79MB -
Statistical Models B/bug_time.txt")
app_time = scan("C:/Users/basil/OneDrive/Desktop/Base/Studies/Undergraduate/Semester 6/F79MB -
Statistical Models B/app_session_duration.txt")

#TASK1A
par(mfrow=c(1,1))

hist(bug_time, #Histogram to visualize distribution shape
     main = "Distribution of Bug Resolution Times",
     xlab = "Resolution Time (Minutes)",
     ylab = "Frequency",
     col = "darkslategray3",
     border = "darkslategray", #Cool color
     breaks = 20)

boxplot(bug_time, #Boxplot with mean highlighted
        main = "Bug Resolution Time Summary",
        horizontal = TRUE,
        xlab = "Minutes",
        col = "aquamarine3",
        border = "aquamarine4",
        pch = 19)

mean_val = mean(bug_time) #Add mean to plot in the box plot to compare with median
points(mean_val, 1,
       pch = 19,
       col = "indianred4",
       cex = 1.5)

legend("topright", #Adding legend to identify its mean
       legend = c("Mean"),
       col = c("indianred4"),
       pch = 19,
       pt.cex = 1.5,
       bty = "n")

summary(bug_time) #Summary including standard deviation and Interquartile range
sd(bug_time)
IQR(bug_time)

#TASK 1Bii
y = log(bug_time)
n = length(y)   #Sample size

#MLE
mhat = mean(y)                      ; mhat
sigma2_mle = sum((y - mhat)^2) / n  ; sigma2_mle
sigma2_sample = var(y)              ; sigma2_sample


#TASK 1C
qqnorm(y, #Q-Q plot to check normality assumption
       main = "Q-Q Plot of Log-Transformed Bug Resolution Times",
       xlab = "Theoretical Normal Quantiles",
       ylab = "Sample Quantiles",
       pch = 19,
       col = "mediumpurple",
       cex = 1.2,
       cex.main = 1.3,
       cex.lab = 1.1,
       cex.axis = 1)

qqline(y, #Add reference line for perfect normality
       col = "mediumpurple4", #Add reference line
       lwd = 2.5,
       lty = 1)

#TASK 1D
#Create 10 equal bins using lognormal quantiles
```

```
breaks = qlnorm(seq(0,1,by=1/10), meanlog=mhat, sdlog=sqrt(sigma2_mle)); breaks #Breaks from lognormal
quantiles

#Cut data into bins
bug.cut = cut(bug_time, breaks=breaks, right=F)
bug.table = table(bug.cut); bug.table

#Extract observed frequencies
obs.f = numeric(10)
for(i in 1:10) obs.f[i] = bug.table[[i]] ; obs.f

#Expected frequencies
exp.f = numeric(10)
for(i in 1:10) exp.f[i] = length(bug_time)*(1/10) ; exp.f

#Calculate chi-square test statistic
chi_sq = sum((obs.f - exp.f)^2 / exp.f) ; chi_sq

df = 10 - 1 - 2 ; df #Degrees of freedom
p_value = 1 - pchisq(chi_sq, df) ; p_value


##############################################################################
#TASK 2A
hist(app_time, #Histogram of original data
     main = "Distribution of App Session Duration",
     xlab = "Resolution Time (Minutes)",
     ylab = "Frequency",
     col = "darkseagreen3",
     border = "darkseagreen4")

boxplot(app_time, #Boxplot with mean highlighted
        main = "App Session Summary",
        horizontal = TRUE,
        xlab = "Minutes",
        col = "aquamarine3",
        border = "aquamarine4")

mean_val = mean(app_time) #Add mean to boxplot
points(mean_val, 1,
       pch = 19,          #Solid circle
       col = "indianred4",   #Orange color
       cex = 1.5)

legend("topright",#Legend for mean
       legend = c("Mean"),
       col = c("indianred4"),
       pch = 19,
       pt.cex = 1.5,
       bty = "n")

#Numerical Summaries
summary(app_time)
sd(app_time)
IQR(app_time)

#TASK2B
B = 100000
n2 = length(app_time)

boot.med = numeric(B)

for(i in 1:B){ #Bootstrap for Median
  samp = sample(app_time, n2, replace=TRUE) #Resample with replacement
  boot.med[i] = median(samp)   #Store median of resample
}


hist(boot.med, #Histogram of bootstrap distribution
     main = "Bootstrap Distribution of Median",
     xlab = "Median (Minutes)",
     ylab = "Frequency",
     col = "khaki2",
```

```
     border = "khaki4",
     breaks = 25)

#Calculate and display 95% CI
ci = quantile(boot.med, c(0.025, 0.975))
abline(v = ci, col = "khaki4", lty = 2)

#Add CI lines to histogram
legend("topright",
       legend = "95% CI",
       col = "khaki4",
       lty = 2)

#Summary statistics of bootstrap distribution
summary(boot.med)
sd(boot.med)
IQR(boot.med)
quantile(boot.med, c(0.025,0.975))

#Perform KS test for normality (even though its not part of our syllabus I still wanted to confirm)
ks_result = ks.test(boot.med, "pnorm",
                    mean = mean(boot.med),
                    sd = sd(boot.med))

#Display results
print(ks_result) #Making sure if its normal because the graph was not sufficient enough

#TASK 2C
log_app = log(app_time) #Log-transform the data
mhat2 = median(log_app); mhat2

#Density at mhat2
f_mhat = dnorm(mhat2, mean = mean(log_app), sd = sd(log_app)); f_mhat

#Standard error of median
se_mhat = 1 / (2 * f_mhat * sqrt(n2)); se_mhat

#95% CI for log-median
lower_mu = mhat2 - 1.96*(sd(log_app)/sqrt(n2)); lower_mu
upper_mu = mhat2 + 1.96*(sd(log_app)/sqrt(n2)); upper_mu

lower_median = exp(lower_mu); lower_median #Transform to median scale
upper_median = exp(upper_mu); upper_median

#TASK 2D
boot.iqr = numeric(B)

for(i in 1:B){ #Bootstrap for IQR
  samp = sample(app_time, n2, replace=TRUE)
  boot.iqr[i] = IQR(samp)
}

quantile(boot.iqr, c(0.025,0.975))

#TASK 2E
mu0 = log(15)
sigma0 = sd(log_app)

boot.param.med = numeric(B)

for(i in 1:B){ #Bootstap for median (another one)
  sim = rlnorm(n2, meanlog=mu0, sdlog=sigma0)
  boot.param.med[i] = median(sim)
}

obs.med = median(app_time)
p_value = mean(boot.param.med >= obs.med); p_value
```

References

[1] *F79MB Project 1 2025 Heriot-Watt University Canvas*. Available at: https://canvas.hw.ac.uk/courses/32228/files/4652596?wrap=1.